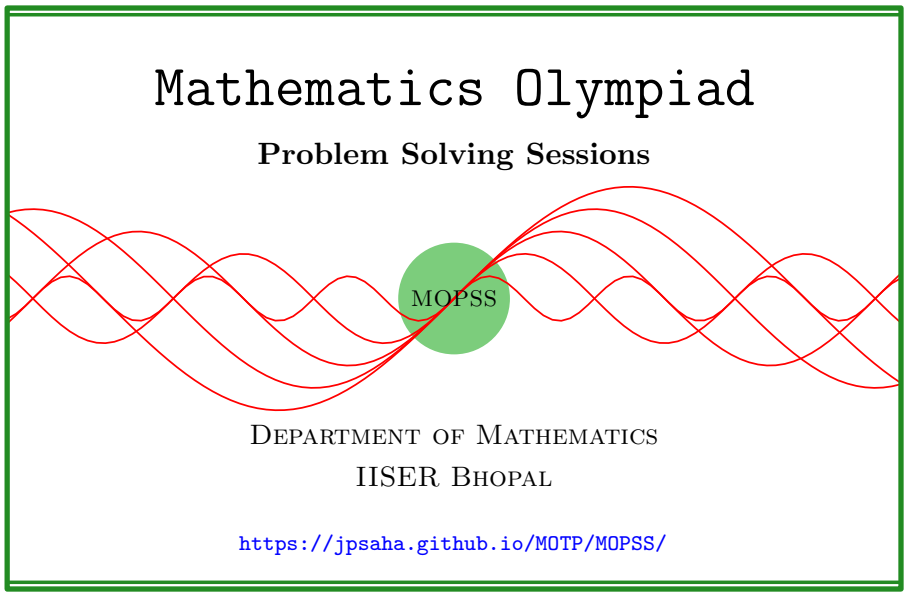
Inequalities

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Inequalities

§1.1 Warm up

Exercise 1 (RMO 1990 P2, , Nesbitt's inequality 1903, Moscow Mathematical Olympiad 1963 Grade 9 Second Round P3, UK BMO 1976 P2)**.** Let

$a, b, c > 0$. Then

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

This has many proofs, for instance using AM-GM, AM-HM, Cauchy-Schwarz inequality, rearrangement inequality. We present a quick proof from [Hun08].

Solution 1. Put

$$\alpha = \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b},$$
$$\beta = \frac{b}{b+c} + \frac{c}{c+a} + \frac{a}{a+b},$$
$$\gamma = \frac{c}{b+c} + \frac{a}{c+a} + \frac{b}{a+b}.$$

By the AM-GM inequality, the inequalities $\alpha + \beta \geq 3, \beta + \gamma \geq 3, \gamma + \alpha \geq 3$ hold. Adding them together yields $2(\alpha + \beta + \gamma) \geq 9$. Using $\beta + \gamma = 3$, we obtain $\alpha \geq 3/2$. ■

§1.2 No square is negative ($x^2 \geq 0$)

Exercise 2 (Canadian Mathematical Olympiad 1971 P2)**.** Prove that if $x > 0, y > 0$ and $x + y = 1$, then

$$\left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right) \geq 9.$$

> **Walkthrough** —
>
> **(a)**

Solution 2. Note that

$$\begin{aligned}\left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right) &= 1 + \frac{1}{x} + \frac{1}{y} + \frac{1}{xy} \\ &= 1 + 1 + \frac{y}{x} + 1 + \frac{x}{y} + \frac{x+y}{xy} \\ &= 1 + 2\left(2 + \frac{x}{y} + \frac{y}{x}\right) \\ &\geq 9.\end{aligned}$$

■

Exercise 3 (All-Russian Mathematical Olympiad 1990,). **.** Prove that for arbitrary $t \in \mathbb{R}$, the inequality $t^4 - t + \frac{1}{2} > 0$ holds.

Walkthrough —
(a)

Solution 3. For any $t \in \mathbb{R}$, note that

$$\begin{aligned} t^4 - t + \frac{1}{2} &= t^4 - t^2 + \frac{1}{4} + t^2 - t + \frac{1}{4} \\ &= \left(t^2 - \frac{1}{2}\right)^2 + \left(t - \frac{1}{2}\right)^2 \geq 0, \end{aligned}$$

where equality occurs only if $t = \frac{1}{2}$ and $t^2 = \frac{1}{2}$, which is impossible. This shows that $t^4 - t + \frac{1}{2} > 0$ for any real number t. ■

Exercise 4 (RMO 1995 P7,). Show that for any real number x:

$$x^2 \sin x + x \cos x + x^2 + \frac{1}{2} > 0.$$

Solution 4. When $1 + \sin x \neq 0$, we have

$$\begin{aligned} & x^2 \sin x + x \cos x + x^2 + \frac{1}{2} \\ &= (1 + \sin x)x^2 + x \cos x + \frac{1}{2} \\ &= (1 + \sin x)\left(x + \frac{\cos x}{2(1 + \sin x)}\right)^2 + \frac{1}{2} - \frac{\cos^2 x}{4(1 + \sin x)} \\ &= (1 + \sin x)\left(x + \frac{\cos x}{2(1 + \sin x)}\right)^2 + \frac{2 + 2\sin x - \cos^2 x}{4(1 + \sin x)} \\ &= (1 + \sin x)\left(x + \frac{\cos x}{2(1 + \sin x)}\right)^2 + \frac{(1 + \sin x)^2}{4} > 0. \end{aligned}$$

If $\sin x = -1$, then $x^2 \sin x + x \cos x + x^2 + \frac{1}{2}$ is equal to $\frac{1}{2}$ and hence it is positive. This completes the proof. ■

Exercise 5 (RMO 2014d P2,). If x and y are positive real numbers, prove that

$$4x^4 + 4y^3 + 5x^2 + y + 1 \geq 12xy.$$

Solution 5. Note that

$$\begin{aligned} 4x^4 + 4y^3 + 5x^2 + y + 1 &\geq (4x^4 + 1) + 5x^2 + (4y^3 + y) \\ &= (2x^2 - 1)^2 + 9x^2 + y(2y - 1)^2 + 4y^2 \end{aligned}$$

$$\begin{aligned}
&\geq 9x^2 + 4y^2 \\
&= (3x - 2y)^2 + 12xy \\
&\geq 12xy.
\end{aligned}$$

■

Exercise 6 (RMO 2014c P2,). Find all real x, y such that

$$x^2 + 2y^2 + \frac{1}{2} \leq x(2y + 1).$$

Solution 6. Let x, y be reals satisfying the above inequality. Note that

$$\begin{aligned}
x^2 + 2y^2 + \frac{1}{2} - x(2y + 1) &= x^2 + 2y^2 + \frac{1}{2} - 2xy - x \\
&= 2(y^2 - xy) + x^2 - x + \frac{1}{2} \\
&= 2\left(y - \frac{x}{2}\right)^2 + \frac{x^2}{2} - x + \frac{1}{2} \\
&= 2\left(y - \frac{x}{2}\right)^2 + \frac{1}{2}(x - 1)^2 \\
&\geq 0.
\end{aligned}$$

For real x, y, the given inequality is equivalent to $x - 1 = 0, y - \frac{x}{2} = 0$, which holds if and only if (x, y) is equal to $(1, \frac{1}{2})$. ■

Exercise 7 (RMO 2015c P7,). Let $x, y, z \in \mathbb{R}$, such that $x^2 + y^2 + z^2 - 2xyz = 1$. Prove that

$$(1 + x)(1 + y)(1 + z) \leq 4 + 4xyz.$$

Solution 7. Note that

$$\begin{aligned}
&4 + 4xyz - (1 + x)(1 + y)(1 + z) \\
&= 4 + 4xyz - (1 + x + y + z + x^2 + y^2 + z^2 + xyz) \\
&= 4 + 3xyz - (1 + x + y + z + x^2 + y^2 + z^2) \\
&= 4 + 3\frac{x^2 + y^2 + z^2 - 1}{2} - (1 + x + y + z + x^2 + y^2 + z^2) \\
&= \frac{1}{2}(5 + 3(x^2 + y^2 + z^2) - 2(1 + x + y + z + x^2 + y^2 + z^2)) \\
&= \frac{1}{2}(3 + x^2 + y^2 + z^2 - 2(x + y + z)) \\
&= \frac{1}{2}((x - 1)^2 + (y - 1)^2 + (z - 1)^2),
\end{aligned}$$

which is nonnegative. So the required inequality follows. ■

Exercise 8 (RMO 2015f P1, Putnam 1998 B1)**.** Find the minimum value of

$$\frac{(x+1/x)^6-(x^6+1/x^6)-2}{(x+1/x)^3+(x^3+1/x^3)}$$

for $x \in \mathbb{R}^+$.

Solution 8. Note that

$$\begin{aligned}\frac{(x+1/x)^6-(x^6+1/x^6)-2}{(x+1/x)^3+(x^3+1/x^3)} &= \frac{(x+1/x)^6-(x^3+1/x^3)^2}{(x+1/x)^3+(x^3+1/x^3)}\\ &= (x+1/x)^3-(x^3+1/x^3)\\ &= 3\left(x+\frac{1}{x}\right)\\ &= 6+3\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2.\end{aligned}$$

It follows that the required minimum value is equal to 6, which is attained at $x=1$. ■

Exercise 9 (RMO 2018a P2,). Let n be a natural number. Find all real numbers x satisfying the equation

$$\sum_{k=1}^{n}\frac{kx^k}{1+x^{2k}}=\frac{n(n+1)}{4}.$$

Solution 9. Let x be a real number satisfying the above equation. It follows that $x \neq 0$. Note that

$$\begin{aligned}\frac{n(n+1)}{4} &= \sum_{k=1}^{n}\frac{kx^k}{1+x^{2k}}\\ &= \sum_{k=1}^{n}\frac{k}{x^k+\frac{1}{x^k}}\\ &\leq \sum_{k=1}^{n}\frac{k}{|x|^k+\frac{1}{|x|^k}}\\ &\leq \sum_{k=1}^{n}\frac{k}{2}\\ &= \frac{n(n+1)}{4}.\end{aligned}$$

Consequently, the inequalities in the intermediate steps are equalities. This shows that x is positive and $|x|=1$, which gives $x=1$. Also note that the given equation holds if $x=1$. Hence, $x=1$ is the only real solution of the given equation. ■

§1.3 Manipulation

Exercise 10 (INMO 1987 P2,). Determine the largest number in the infinite sequence

$$1, 2^{1/2}, 3^{1/3}, 4^{1/4}, \ldots, n^{1/n}, \ldots.$$

Solution 10. Note that $1 < 2^{1/2} < 3^{1/3}$ holds. Let us establish the following Claim.

> **Claim —** For any integer $n \geq 3$, the inequality
>
> $$n^{\frac{1}{n}} > (n+1)^{\frac{1}{n+1}}$$
>
> holds.

Proof of the Claim. Observe that the inequality is equivalent to $n > \left(1+\frac{1}{n}\right)^n$. For any integer $n \geq 3$, note that

$$\begin{aligned}
&\left(1+\frac{1}{n}\right)^n \\
&= 1 + \binom{n}{1}\frac{1}{n} + \binom{n}{2}\frac{1}{n^2} + \cdots + \binom{n}{n}\frac{1}{n^n} \\
&= 1 + \frac{n}{1!}\frac{1}{n} + \frac{n(n-1)}{2!}\frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!}\frac{1}{n^3} \\
&\quad + \cdots + \frac{n(n-1)(n-2)\cdots 2}{(n-1)!}\frac{1}{n^{n-1}} + \frac{n(n-1)(n-2)\ldots 1}{n!}\frac{1}{n^n} \\
&= 1 + \frac{1}{1!} + \frac{1}{2!}\left(1-\frac{1}{n}\right) + \frac{1}{3!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) + \\
&\quad \cdots + \frac{1}{(n-1)!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{n-2}{n}\right) \\
&\qquad + \frac{1}{n!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{n-2}{n}\right)\left(1-\frac{n-1}{n}\right) \\
&< 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots + \frac{1}{n!} \\
&< 2 + \frac{1}{2} + \frac{1}{2\cdot 2} + \frac{1}{2\cdot 2\cdot 2} + \cdots + \frac{1}{\underbrace{2\cdot 2\cdot 2\cdot\cdots\cdot 2}_{(n-1)\text{-times}}} \\
&= 2 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n-1}} \\
&< 3 \\
&\leq n
\end{aligned}$$

holds. This proves the Claim. □

The content posted here and at this blog by Evan Chen are quite useful.

It follows that the largest term in the given sequence is equal to $3^{1/3}$. ■

Remark. For any $n \geq e$, note that

$$n^{1/n} \geq e^{1/n} > 1 + \frac{1}{n}$$

holds, which shows that $n^{1/n} > (n+1)^{1/(n+1)}$.

Example 11. Suppose $(x_1, x_2, \cdots, x_n, \cdots)$ is a sequence of positive real numbers such that $x_1 \geq x_2 \geq x_3 \geq \cdots x_n \geq \cdots$, and for all n,

$$\frac{x_1}{1} + \frac{x_4}{2} + \frac{x_9}{3} + \cdots + \frac{x_{n^2}}{n} \leq 1.$$

Show that for all $k \geq 1$ the following inequality is satisfied:

$$\frac{x_1}{1} + \frac{x_2}{2} + \frac{x_3}{3} + \cdots + \frac{x_k}{k} \leq 3.$$

Solution 11. Note that for any integer $n \geq 1$, we have

$$\begin{aligned}
&\left(\frac{x_1}{1} + \frac{x_2}{2} + \frac{x_3}{3}\right) \\
&+ \left(\frac{x_4}{4} + \frac{x_5}{5} + \frac{x_6}{6} + \frac{x_7}{7} + \frac{x_8}{8}\right) \\
&\quad + \cdots + \left(\frac{x_{n^2}}{n^2} + \frac{x_{n^2+1}}{n^2+1} + \cdots + \frac{x_{(n+1)^2-1}}{(n+1)^2-1}\right) \\
&\leq \left(\frac{x_1}{1} + \frac{x_1}{1} + \frac{x_1}{1}\right) + \left(\frac{x_4}{4} + \frac{x_4}{4} + \frac{x_4}{4} + \frac{x_4}{4} + \frac{x_4}{4}\right) \\
&+ \cdots + \underbrace{\left(\frac{x_{n^2}}{n^2} + \frac{x_{n^2}}{n^2} + \cdots + \frac{x_{n^2}}{n^2}\right)}_{(2n+1) \text{ terms}} \\
&= (2 \cdot 1 + 1)\frac{x_1}{1} + (2 \cdot 2 + 1)\frac{x_4}{4} + \cdots + (2n+1)\frac{x_{n^2}}{n^2} \\
&\leq (3 \cdot 1)\frac{x_1}{1} + (3 \cdot 2)\frac{x_4}{4} + \cdots + (3n)\frac{x_{n^2}}{n^2} \\
&= 3\left(\frac{x_1}{1} + \frac{x_4}{2} + \cdots + \frac{x_{n^2}}{n}\right) \\
&\leq 3.
\end{aligned}$$

Since for any $k \geq 1$, there is a positive integer m with $(m+1)^2 - 1 \geq k$, the result follows. ■

Example 12. For any natural number $n > 1$, prove the inequality

$$\frac{1}{2} < \frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + \cdots + \frac{n}{n^2+n} < \frac{1}{2} + \frac{1}{2n}.$$

Solution 12. Note that

$$\begin{aligned}&\frac{1}{n^2+1}+\frac{2}{n^2+2}+\frac{3}{n^2+3}+\cdots+\frac{n}{n^2+n}\\&>\frac{1}{n^2+n}(1+2+3+\cdots+n)\quad(\text{using } n>1)\\&=\frac{1}{2}.\end{aligned}$$

Also note that

$$\begin{aligned}&\frac{1}{n^2+1}+\frac{2}{n^2+2}+\frac{3}{n^2+3}+\cdots+\frac{n}{n^2+n}\\&<\frac{1}{n^2+1}(1+2+3+\cdots+n)\quad(\text{using } n>1)\\&=\frac{n^2+n}{2(n^2+1)}\\&=\frac{1}{2}+\frac{n-1}{2(n^2+1)}\\&<\frac{1}{2}+\frac{1}{2n}.\end{aligned}$$

■

Example 13. If a,b,c are three real numbers such that $|a-b|\geq|c|, |b-c|\geq|a|, |c-a|\geq|b|$, then prove that one of a,b,c is the sum of the other two.

Solution 13. The given inequalities are equivalent to $(a-b)^2-c^2\geq 0, (b-c)^2-a^2\geq 0, (c-a)^2-b^2\geq 0$, which yields

$$\begin{aligned}(a-b+c)(a-b-c)&\geq 0,\\(b-c+a)(b-c-a)&\geq 0,\\(c-a+b)(c-a-b)&\geq 0.\end{aligned}$$

Multiplying them, we obtain

$$-(b+c-a)^2(c+a-b)^2(a+b-c)^2\geq 0,$$

which shows that $(b+c-a)(c+a-b)(a+b-c)$ is equal to 0. This proves the result. ■

§1.4 Rearrangement inequality

Theorem 1 (Rearrangement inequality)

Let $a_1, \ldots, a_n, b_1, \ldots, b_n$ be real numbers satisfying $a_1 \geq a_2 \geq \cdots \geq a_n, b_1 \geq b_2 \geq \cdots \geq b_n$. Then for any permutation σ of $\{1, 2, \ldots, n\}$,

$$\begin{aligned} a_1b_1 + a_2b_2 + \cdots + a_nb_n &\geq a_1b_{\sigma(1)} + a_2b_{\sigma(2)} + \cdots + a_nb_{\sigma(n)} \\ &\geq a_1b_n + a_2b_{n-1} + \cdots + a_nb_1 \end{aligned}$$

holds. In other words, for any two sequences $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ of real numbers, the sum $a_1b_1 + \cdots + a_nb_n$ is maximized (resp. minimized) when these sequences are sorted in the same (resp. opposite) order.

Exercise 14 (Canadian Mathematical Olympiad 2002 P3,). For positive $x, y, z \in \mathbb{R}$, prove that

$$\frac{x^3}{yz} + \frac{y^3}{zx} + \frac{z^3}{xy} \geq x + y + z.$$

Walkthrough —

(a)

Solution 14. Since x, y, z are positive, it follows that the sequences (x^3, y^3, z^3) and $(\frac{1}{yz}, \frac{1}{zx}, \frac{1}{xy})$ are similarly ordered. By the rearrangement inequality, it follows that

$$\frac{x^3}{yz} + \frac{y^3}{zx} + \frac{z^3}{xy} \geq \frac{x^3}{zx} + \frac{y^3}{xy} + \frac{z^3}{yz} = \frac{x^2}{z} + \frac{y^2}{x} + \frac{z^2}{y}.$$

Since the sequences (x^2, y^2, z^2) and $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$ are sorted in the opposite order, using the rearrangement inequality, we get

$$\frac{x^2}{z} + \frac{y^2}{x} + \frac{z^2}{y} \geq \frac{x^2}{x} + \frac{y^2}{y} + \frac{z^2}{z} = x + y + z.$$

Combining the above inequalities, we obtain

$$\frac{x^3}{yz} + \frac{y^3}{zx} + \frac{z^3}{xy} \geq x + y + z.$$

■

Example 15. For positive reals a, b, c, show that $a^7+b^7+c^7 \geq a^4b^3+b^4c^3+c^4a^3$.

Solution 15. Since a, b, c are positive reals, it follows that the sequences a^4, b^4, c^4 and a^3, b^3, c^3 are sorted in the same order. By the rearrangement inequality, we obtain

$$a^7 + b^7 + c^7 \geq a^4b^3 + b^4c^3 + c^4a^3.$$

■

Exercise 16 (RMO 2012a P3, , RMO 2012b P3, RMO 2012c P3, RMO 2012d P3)**.** Let a and b be positive real numbers such that $a + b = 1$. Prove that $a^ab^b + a^bb^a \leq 1$.

Solution 16. For any positive a, b, the sequences $(a^a, b^a), (a^b, b^b)$ are sorted the same way. Applying the rearrangement inequality, we obtian

$$a^ab^b + a^bb^a \leq a^aa^b + b^ab^b = a^{a+b} + b^{a+b} = a + b = 1.$$

■

Exercise 17 (RMO 2012b P3, , RMO 2012a P3, RMO 2012c P3, RMO 2012d P3)**.** Let a and b be positive real numbers such that $a + b = 1$. Prove that $a^ab^b + a^bb^a \leq 1$.

Solution 17. For any positive a, b, the sequences $(a^a, b^a), (a^b, b^b)$ are sorted the same way. Applying the rearrangement inequality, we obtian

$$a^ab^b + a^bb^a \leq a^aa^b + b^ab^b = a^{a+b} + b^{a+b} = a + b = 1.$$

■

Exercise 18 (RMO 2012c P3, , RMO 2012a P3, RMO 2012b P3, RMO 2012d P3)**.** Let a and b be positive real numbers such that $a + b = 1$. Prove that $a^ab^b + a^bb^a \leq 1$.

Solution 18. For any positive a, b, the sequences $(a^a, b^a), (a^b, b^b)$ are sorted the same way. Applying the rearrangement inequality, we obtian

$$a^ab^b + a^bb^a \leq a^aa^b + b^ab^b = a^{a+b} + b^{a+b} = a + b = 1.$$

■

Exercise 19 (RMO 2012d P3, , RMO 2012a P3, RMO 2012b P3, RMO 2012c P3)**.** Let a and b be positive real numbers such that $a + b = 1$. Prove that $a^ab^b + a^bb^a \leq 1$.

Solution 19. For any positive a, b, the sequences $(a^a, b^a), (a^b, b^b)$ are sorted the same way. Applying the rearrangement inequality, we obtian

$$a^a b^b + a^b b^a \leq a^a a^b + b^a b^b = a^{a+b} + b^{a+b} = a + b = 1.$$

■

Exercise 20 (RMO 2014b P2,). Let x, y, z be positive real numbers. Prove that

$$\frac{y^2+z^2}{x} + \frac{z^2+x^2}{y} + \frac{x^2+y^2}{z} \geq 2(x+y+z).$$

Solution 20. Note that

$$\frac{y^2+z^2}{x} + \frac{z^2+x^2}{y} + \frac{x^2+y^2}{z} = \left(\frac{y^2}{x} + \frac{z^2}{y} + \frac{x^2}{z}\right) + \left(\frac{z^2}{x} + \frac{x^2}{y} + \frac{y^2}{z}\right).$$

holds. Since (x^2, y^2, z^2) and $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$ are sorted oppositely, by the rearrangement inequality, we obtain

$$\left(\frac{y^2}{x} + \frac{z^2}{y} + \frac{x^2}{z}\right) \geq x + y + z, \quad \left(\frac{z^2}{x} + \frac{x^2}{y} + \frac{y^2}{z}\right) \geq x + y + z.$$

Adding these inequalities, the required inequality follows. ■

> **Remark.** Note that the last two inequalities also follow from the Cauchy–Schwarz inequality.

Exercise 21 (INMO 2001 P3,). If a, b, c are positive real numbers such that $abc = 1$, Prove that

$$a^{b+c} b^{c+a} c^{a+b} \leq 1.$$

Solution 21. Put

$$\alpha = a^{b+c} b^{c+a} c^{a+b},$$
$$\beta = b^{b+c} c^{c+a} a^{a+b},$$
$$\gamma = c^{b+c} a^{c+a} b^{a+b}.$$

Note that $\alpha\beta\gamma = 1$, and by the rearrangement inequality, it follows that

$$\alpha \leq \beta, \alpha \leq \gamma$$

hold. This gives $\alpha^3 \leq 1$, which yields $\alpha \leq 1$. ■

Remark. The above proof is similar to be the proof of Nesbitt's inequality as in [Hun08].

Exercise 22 (RMO 2017a P6,). Let x, y, z be real numbers, each greater than 1. Prove that

$$\frac{x+1}{y+1}+\frac{y+1}{z+1}+\frac{z+1}{x+1} \leq \frac{x-1}{y-1}+\frac{y-1}{z-1}+\frac{z-1}{x-1}$$

Solution 22. Note that both sides of the above inequality remain invariant under cyclic permutations of x, y, z. Therefore, without loss of generality, we assume that $x \geq y$ and $x \geq z$. Note that

$$\begin{aligned}\frac{x-1}{y-1}-\frac{x+1}{y+1} &= \frac{(x-1)(y+1)-(x+1)(y-1)}{y^2-1} \\ &= \frac{2(x-y)}{y^2-1}.\end{aligned}$$

Similarly, it follows that

$$\begin{aligned}\frac{y-1}{z-1}-\frac{y+1}{z+1} &= \frac{2(y-z)}{z^2-1}, \\ \frac{z-1}{x-1}-\frac{z+1}{x+1} &= \frac{2(z-x)}{x^2-1}.\end{aligned}$$

It suffices to prove that

$$\frac{x-y}{y^2-1}+\frac{y-z}{z^2-1}+\frac{z-x}{x^2-1} \geq 0,$$

which is equivalent to

$$\frac{x}{y^2-1}+\frac{y}{z^2-1}+\frac{z}{x^2-1} \geq \frac{x}{x^2-1}+\frac{y}{y^2-1}+\frac{z}{z^2-1},$$

which follows from the rearrangement inequality since x, y, z and $\frac{1}{x^2-1}, \frac{1}{y^2-1}, \frac{1}{z^2-1}$ are sorted in the opposite order. ■

§1.5 Cauchy–Schwarz inequality

Exercise 23 (Lagrange's identity). Prove that

$$\begin{aligned}&(a_1^2+a_2^2+\cdots+a_n^2)(b_1^2+b_2^2+\cdots+b_n^2) \\ &= \sum_{1 \leq i<j \leq n}(a_ib_j-a_jb_i)^2\end{aligned}$$

for all real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$.

The content posted here and at this blog by Evan Chen are quite useful.

Exercise 24 (All-Russian Mathematical Olympiad 2018 Grade 11 Day 1 P2, , by F. Petrov)**.** Let $n \geq 2$ and $x_1, x_2, \ldots, x_n$ be positive real numbers. Prove that

$$\frac{1+x_1^2}{1+x_1x_2} + \frac{1+x_2^2}{1+x_2x_3} + \cdots + \frac{1+x_n^2}{1+x_nx_1} \geq n.$$

Walkthrough —

(a) Use the Cauchy–Schwarz inequality to obtain a lower bound for each term.

(b) Use the AM–GM inequality to complete the proof.

Solution 23. Note that

$$(1+a^2)(1+b^2) \geq (1+ab)^2$$

holds for any real numbers a, b by the Cauchy–Schwarz inequality. This implies that

$$\begin{aligned}
&\frac{1+x_1^2}{1+x_1x_2} + \frac{1+x_2^2}{1+x_2x_3} + \cdots + \frac{1+x_n^2}{1+x_nx_1} \\
&\geq \sqrt{\frac{1+x_1^2}{1+x_2^2}} + \sqrt{\frac{1+x_2^2}{1+x_3^2}} + \cdots + \sqrt{\frac{1+x_n^2}{1+x_1^2}} \\
&\geq n,
\end{aligned}$$

where the last inequality follows from the AM–GM inequality. This completes the proof. ■

Exercise 25 (Bay Area MO 2000 P3,)**.** Let $x_1, x_2, \ldots, x_n$ be positive numbers, with $n \geq 2$. Prove that

$$\left(x_1 + \frac{1}{x_1}\right)\left(x_2 + \frac{1}{x_2}\right)\ldots\left(x_n + \frac{1}{x_n}\right) \geq \left(x_1 + \frac{1}{x_2}\right)\left(x_2 + \frac{1}{x_3}\right)\ldots\left(x_n + \frac{1}{x_1}\right).$$

Walkthrough —

(a)

Solution 24.

■

Example 26. Show that if the sum of positive numbers a, b, c is equal to 1, then

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9.$$

Solution 25. Applying the Cauchy–Schwarz inequality, we obtain

$$(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)\geq 3^2.$$

Using $a+b+c=1$, the required inequality follows. ■

Exercise 27 (RMO 2013d P3,). Given real numbers $a,b,c,d,e>1$. Prove that

$$\frac{a^2}{c-1}+\frac{b^2}{d-1}+\frac{c^2}{e-1}+\frac{d^2}{a-1}+\frac{e^2}{b-1}\geq 20.$$

Solution 26. Since $a-1,b-1,c-1,d-1,e-1$ are all positive, by applying the Cauchy-Schwarz inequality, we get

$$\frac{a^2}{c-1}+\frac{b^2}{d-1}+\frac{c^2}{e-1}+\frac{d^2}{a-1}+\frac{e^2}{b-1}\geq\frac{(a+b+c+d+e)^2}{a+b+c+d+e-5}.$$

It suffices to show that

$$\frac{(a+b+c+d+e)^2}{a+b+c+d+e-5}\geq 20$$

holds. Since $a+b+c+d+e-5$ is positive, it is enough to prove that

$$(a+b+c+d+e)^2-20(a+b+c+d+e)+100\geq 0,$$

which holds

$$(a+b+c+d+e)^2-20(a+b+c+d+e)+100=(a+b+c+d+e-10)^2.$$

This proves the result. ■

Exercise 28 (RMO 2014a P6,). Let $x_1,x_2,x_3,\ldots,x_{2014}$ be positive real numbers such that $\sum_{j=1}^{2014}x_j=1$. Determine with proof the smallest constant K such that

$$K\sum_{j=1}^{2014}\frac{x_j^2}{1-x_j}\geq 1. \tag{1}$$

Solution 27. For any positive reals $x_1,\ldots,x_{2014}$ satisfying $\sum_{j=1}^{2014}x_j=1$, using the Cauchy–Schwarz inequality, we obtain

$$\sum_{j=1}^{2014}\frac{x_j^2}{1-x_j}\geq\frac{(x_1+\cdots+x_{2014})^2}{2014-(x_1+\cdots+x_{2014})}=\frac{1}{2013},$$

which gives

$$2013\sum_{j=1}^{2014}\frac{x_j^2}{1-x_j}\geq 1.$$

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This shows that the inequality (??) holds for $K = 2013$.

For $x_1 = x_2 = \cdots = x_{2014} = \frac{1}{2014}$, note that

$$\sum_{j=1}^{2014} \frac{x_j^2}{1 - x_j} = \frac{1}{2013}$$

holds, which shows that for any K satisfying the inequality (??), the bound $K \geq 2013$ holds.

This proves that the smallest constant K satisfying the inequality (??) is equal to 2013. ■

Exercise 29 (RMO 2016d P2,). Let a, b, c be positive real numbers such that

$$\frac{ab}{1+bc} + \frac{bc}{1+ca} + \frac{ca}{1+ab} = 1.$$

Prove that

$$\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} \geq 6\sqrt{2}.$$

Solution 28. Applying the Cauchy–Schwarz inequality, we obtain

$$\begin{aligned} 1 &= \frac{ab}{1+bc} + \frac{bc}{1+ca} + \frac{ca}{1+ab} \\ &\geq \frac{(\sqrt{ab} + \sqrt{bc} + \sqrt{ca})^2}{3 + ab + bc + ca}, \end{aligned}$$

which yields

$$3 \geq 2\sqrt{abc}(\sqrt{a} + \sqrt{b} + \sqrt{c}).$$

Using the AM-GM inequality, we get

$$2\sqrt{abc}\sqrt[3]{\sqrt{abc}} \leq 1,$$

which gives $abc \leq \frac{1}{2\sqrt{2}}$. Note that

$$\begin{aligned} \frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} &\geq \frac{3}{abc} \\ &\geq 6\sqrt{2}. \end{aligned}$$

■

We refer to Exercise 62 and Exercise 63.

Exercise 30 (RMO 2023b P5,). Let $n > k > 1$ be positive integers. Determine all positive real numbers $a_1, a_2, \ldots, a_n$ which satisfy

$$\sum_{i=1}^{n} \sqrt{\frac{ka_i^k}{(k-1)a_i^k + 1}} = \sum_{i=1}^{n} a_i = n.$$

Solution 29. Let $a_1, \ldots, a_n$ be positive reals satisfying the above condition. Note that

$$\begin{aligned}
&\sum_{i=1}^{n} \sqrt{\frac{ka_i^k}{(k-1)a_i^k+1}} \\
&\leq \sum_{i=1}^{n} \sqrt{\frac{a_i^k}{\sqrt[k]{a_i^{k(k-1)}}}} \quad \text{(by the AM-GM inequality)} \\
&= \sum_{i=1}^{n} \sqrt{a_i} \\
&\leq \left(n \sum_{i=1}^{n} a_i \right)^{1/2} \quad \text{(by the Cauchy–Schwarz inequality)} \\
&= \sum_{i=1}^{n} \sqrt{\frac{ka_i^k}{(k-1)a_i^k+1}}
\end{aligned}$$

hold, and hence, the intermediate inequalities are equalities. This implies that $a_1, a_2, \ldots, a_n$ are equal. Using the given condition, it follows that

$$a_1 = a_2 = \cdots = a_n = 1.$$

Also note that the given conditions are satisfied if $a_1 = a_2 = \cdots = a_n = 1$ holds. This proves that the positive real numbers satisfying the given condition are precisely

$$a_1 = a_2 = \cdots = a_n = 1.$$

■

§1.6 QM-AM-GM-HM inequality

Example 31. Prove that if $m > 0$, then

$$m + \frac{4}{m^2} \geq 3.$$

Solution 30. Applying the AM-GM inequality, it follows that

$$m + \frac{4}{m^2} = \frac{m}{2} + \frac{m}{2} + \frac{4}{m^2} \geq 3.$$

■

Exercise 32 (INMO 1988 P4,). If $a, b > 0$ with $a + b = 1$, then show that $(a + \frac{1}{a})^2 + (b + \frac{1}{b})^2 \geq \frac{25}{2}$.

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Solution 31. Note that

$$\begin{aligned}\left(a+\frac{1}{a}\right)^2+\left(b+\frac{1}{b}\right)^2 &= a^2+b^2+\frac{1}{a^2}+\frac{1}{b^2}+4\\ &= a^2+b^2+1+\frac{b^2}{a^2}+2\frac{b}{a}+1+\frac{a^2}{b^2}+2\frac{a}{b}+4\\ &= a^2+b^2+\left(\frac{a}{b}-\frac{b}{a}\right)^2+\left(\sqrt{\frac{a}{b}}-\sqrt{\frac{b}{a}}\right)^2+12\\ &\geq \frac{(a+b)^2}{2}+12\\ &= \frac{25}{2}.\end{aligned}$$

■

Exercise 33 (RMO 1991 P2,). If a, b, c and d are any 4 positive real numbers, then prove that

$$\frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{a}\geq 4.$$

Solution 32. Note that

$$\begin{aligned}\frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{a} &\geq 2\sqrt{\frac{a}{c}}+2\sqrt{\frac{c}{a}}\\ &\geq 4.\end{aligned}$$

■

Exercise 34 (RMO 1994 P8, , All-Russian MO 1991 Grade 11 Day 1 P3). If a, b and c are positive real numbers such that $a+b+c=1$, prove that

$$(1+a)(1+b)(1+c)\geq 8(1-a)(1-b)(1-c).$$

Solution 33. The given inequality is equivalent to

$$(b+c+2a)(c+a+2b)(a+b+2c)\geq 8(b+c)(c+a)(a+b).$$

Put $x=b+c, y=c+a, z=a+b$. Note that the above inequality can be rewritten as

$$(y+z)(z+x)(x+y)\geq 8xyz,$$

which follows from the AM-GM inequality. ■

Exercise 35 (RMO 1993 P6,). If a, b, c, d are four positive real numbers such that $abcd = 1$, prove that

$$(1 + a)(1 + b)(1 + c)(1 + d) \geq 16.$$

Solution 34. It follows by applying the AM-GM inequality to the factors and using $abcd = 1$. ■

Example 36. Show that if $a, b, c \geq 0$, then

$$ab(a + b) + bc(b + c) + ca(c + a) \geq 6abc.$$

Example 37. [PK74, Problem 62.2] Each of the four numbers a, b, c, and d is positive and less than one. Show that not all four products

$$4a(1 - b), 4b(1 - c), 4c(1 - d), 4d(1 - a)$$

are greater than one.

Solution 35. Note that their product is equal to

$$\prod_{x\in\{a,b,c,d\}} 4x(1 - x) = \prod_{x\in\{a,b,c,d\}} (1 - (2x - 1)^2)$$

which is at most 1. This shows that not all the above four products are greater than one. ■

Exercise 38 (RMO 2003 P3,). Let a, b, c be three positive real numbers such that $a+b+c = 1$. Prove that among the three numbers $a-ab, b-bc, c-ca$, there is one which is at most $\frac{1}{4}$ and there is one which is at least $\frac{2}{9}$.

Solution 36. Note that

$$(a - ab)(b - bc)(c - ca) = a(1 - a)b(1 - b)c(1 - c) \leq \frac{1}{4^3}$$

holds. So among $a - ab, b - bc, c - ca$ there is one which is at most $\frac{1}{4}$.

Also note that

$$\begin{aligned}(a - ab) + (b - bc) + (c - ca) &= 1 - (ab + bc + ca)\\ &\geq 1 - \frac{1}{3}(a + b + c)^2\\ &\geq \frac{2}{3}\end{aligned}$$

holds. This shows that one of $a - ab, b - bc, c - ca$ is at least $\frac{2}{9}$. ■

Exercise 39 (RMO 2005 P7,). Let a, b, c be three positive real numbers such that $a + b + c = 1$. Let

$$\lambda = \min\{a^3 + a^2bc, b^3 + ab^2c, c^3 + abc^2\}.$$

Prove that the roots of the equation $x^2 + x + 4\lambda = 0$ are real.

Solution 37. We need to show that the discriminant of $x^2 + x + 4\lambda$ is nonnegative, that is, $\lambda \leq \frac{1}{16}$ holds. On the contrary, let us assume that $\lambda > \frac{1}{16}$. This gives

$$a^3 + a^2bc > \frac{1}{16}, b^3 + ab^2c > \frac{1}{16}, c^3 + abc^2 > \frac{1}{16}.$$

Note that

$$a^3 + a^2bc = a^2(a + bc) = a^2(1 - b - c + bc) = a^2(1 - b)(1 - c) > \frac{1}{16}$$

holds. Similarly, it follows that

$$b^2(1 - c)(1 - a) > \frac{1}{16}, c^2(1 - a)(1 - b) > \frac{1}{16}.$$

This implies that

$$(abc(1 - a)(1 - b)(1 - c))^2 > \frac{1}{16^3},$$

which is impossible since

$$a(1 - a) \leq \frac{1}{4}, b(1 - b) \leq \frac{1}{4}, c(1 - c) \leq \frac{1}{4}.$$

Consequently, we obtain $\lambda \leq \frac{1}{16}$, and hence the roots of the equation $x^2 + x + 4\lambda = 0$ are real. ■

Example 40. If a, b, c are positive reals, then show that

$$(a^2 + 1)(b^2 + 1)(c^2 + 1) \geq (a + b)(b + c)(c + a).$$

Solution 38. Note that

$$(a^2 + 1)(b^2 + 1) - (a + b)^2 = (ab - 1)^2 \geq 0$$

holds. Similarly, it follows that

$$(b^2 + 1)(c^2 + 1) \geq (b + c)^2, (c^2 + 1)(a^2 + 1) \geq (c + a).$$

Multiplying the above inequalities and using that a, b, c are nonnegative, the required inequality follows. ■

Exercise 41 (RMO 2006 P3,). If a, b, c are three positive real numbers, prove that

$$\frac{a^2+1}{b+c}+\frac{b^2+1}{c+a}+\frac{c^2+1}{a+b}\geq 3.$$

Solution 39. Note that

$$\frac{a^2+1}{b+c}+\frac{b^2+1}{c+a}+\frac{c^2+1}{a+b}\geq \frac{2a}{b+c}+\frac{2b}{c+a}+\frac{2c}{a+b}\geq 3,$$

where the first inequality follows from AM-GM inequality and the second inequality follows from Nesbitt's inequality **??**. ■

Exercise 42 (RMO 2007 P6,). Prove that

(a) $5<\sqrt{5}+\sqrt[3]{5}+\sqrt[4]{5}$,

(b) $8>\sqrt{8}+\sqrt[3]{8}+\sqrt[4]{8}$,

(c) $n>\sqrt{n}+\sqrt[3]{n}+\sqrt[4]{n}$ for all integers $n\geq 9$.

Solution 40. Note that

$$\sqrt{5}>2,\ \sqrt[3]{5}>\frac{8}{5},\ \sqrt[4]{5}>\frac{7}{5}$$

holds, which gives the first inequality. Using $3>\sqrt{8}, 3>\sqrt[4]{8}$, we obtain the second inequality. Note that

$$\sqrt{n}+\sqrt[3]{n}+\sqrt[4]{n}<\sqrt{n}+\sqrt{n}+\sqrt{n}=3\sqrt{n}$$

holds for any integer $n>1$. Since $n\geq 3\sqrt{n}$ holds for any $n\geq 9$, the third inequality follows. ■

Remark. One may also use the AM-GM inequality to obtain the first inequality.

Exercise 43 (RMO 2008 P3,). Suppose a and b are real numbers such that the roots of the cubic equation ax^3-x^2+bx-1 are positive real numbers. Prove that

(i) $0<3ab\leq 1$,

(ii) $b\geq\sqrt{3}$.

Solution 41. Let α, β, γ denote the roots of the polynomial ax^3-x^2+bx-1. Note that

$$\alpha+\beta+\gamma=\frac{1}{a},$$

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$$\alpha\beta+\beta\gamma+\gamma\alpha=\frac{b}{a},$$
$$\alpha\beta\gamma=\frac{1}{a}$$

holds. Since α,β,γ are positive, it follows that a,b are positive. Using

$$(\alpha+\beta+\gamma)^2\geq 3(\alpha\beta+\beta\gamma+\gamma\alpha),$$

we obtain $\frac{1}{a^2}\geq\frac{3b}{a}$. Since a is positive, it follows that $3ab\leq 1$. Note that

$$\begin{aligned}(\alpha\beta+\beta\gamma+\gamma\alpha)^2&=\alpha^2\beta^2+\beta^2\gamma^2+\gamma^2\alpha^2+2\alpha\beta\gamma(\alpha+\beta+\gamma)\\&\geq\frac{(\alpha\beta+\beta\gamma+\gamma\alpha)^2}{3}+2\alpha\beta\gamma(\alpha+\beta+\gamma)\end{aligned}$$

holds, we get $\frac{b^2}{a^2}\geq\frac{b^2}{3a^2}+2\frac{1}{a^2}$, which gives $b^2\geq 3$. Since b is positive, we conclude $b\geq\sqrt{3}$. ■

Exercise 44 (RMO 2011b P6,). Find the largest real constant λ such that

$$\frac{\lambda abc}{a+b+c}\leq(a+b)^2+(a+b+4c)^2 \tag{2}$$

for all positive real numbers a,b,c.

Solution 42. Let λ be a nonnegative real number. Then the above inequality holds for all $a,b,c>0$ if and only if

$$\lambda x^2c\leq(2x+c)(4x^2+4(x+2c)^2)$$

holds for all $x,c>0$. Noting that both the sides are homogeneous of degree three in x and y, it follows that the above inequality holds if and only if

$$\lambda x^2\leq(2x+1)(4x^2+4(x+2)^2)$$

holds for all $x>0$. Observe that

$$\begin{aligned}(2x+1)(4x^2+4(x+2)^2)&=8(2x+1)(x^2+2x+2)\\&=8(2x^3+5x^2+6x+2).\end{aligned}$$

This shows that for $x>0$, the above inequality is equaivalent to

$$\frac{1}{2}\left(\frac{\lambda}{8}-5\right)\leq x+\frac{3}{x}+\frac{1}{x^2}.$$

Let $a,b>0$ be such that

$$x+\frac{3}{x}+\frac{1}{x^2}=ax+\frac{3}{x}+2bx+\frac{1}{x^2}$$

holds for any $x > 0$, and $(3/a)^3 = (1/b)^2$ holds, or equivalently, a, b satisfy $a + 2b = 1$ and $(3/a)^3 = (1/b)^2$. Note that this holds for $a = 3/4, b = 1/8$. Observe that

$$x + \frac{3}{x} + \frac{1}{x^2} = \frac{3}{4}x + \frac{3}{x} + \frac{1}{8}x + \frac{1}{8}x + \frac{1}{x^2} \geq 3 + \frac{3}{4} = \frac{15}{4},$$

where equality holds if and only if $x = 2$. This proves that the largest real constant λ satisfying the given inequality for all $a, b, c > 0$ satisfies

$$\frac{1}{2}\left(\frac{\lambda}{8} - 5\right) = \frac{15}{4},$$

or equivalently, $\lambda = 100$ holds. ■

Exercise 45 (RMO 2012e P4,). Let a, b, c be positive real numbers such that $abc(a + b + c) = 3$. Prove that we have

$$(a + b)(b + c)(c + a) \geq 8.$$

Also determine the case of equality.

Solution 43. Note that

$$\begin{aligned}(a + b)(b + c)(c + a) &= (a + b + c)(ab + bc + ca) - abc \\ &= abc(a + b + c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) - abc \\ &= 3\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) - abc \\ &\geq 9\frac{1}{\sqrt[3]{abc}} - abc.\end{aligned}$$

Also note that

$$\begin{aligned}3 &= abc(a + b + c) \\ &\geq 3(abc)^{4/3},\end{aligned}$$

which gives $abc \leq 1$. Using the above inequalities, we obtain

$$\begin{aligned}(a + b)(b + c)(c + a) &\geq 9 - 1 \\ &= 8.\end{aligned}$$

This proves that

$$(a + b)(b + c)(c + a) \geq 8,$$

where equality holds if and only if all the prior inequalities are equalities, or equivalently, $a = b = c$. Using $abc(a+b+c) = 3$, it follows that a, b, c are equal if and only if they are equal to 1. This shows that $(a + b)(b + c)(c + a) = 8$ if and only if a, b, c are equal to 1. ■

Exercise 46 (RMO 2012f P8,). Let x, y, z be positive real numbers such that $2(xy + yz + zx) = xyz$. Prove that

$$\frac{1}{(x-2)(y-2)(z-2)} + \frac{8}{(x+2)(y+2)(z+2)} \leq \frac{1}{32}.$$

If a, b, c are positive real numbers such that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$, then show that

$$\frac{1}{(a-1)(b-1)(c-1)} + \frac{8}{(a+1)(b+1)(c+1)} \leq \frac{1}{4}.$$

Solution 44. Put

$$x = 2a, y = 2b, z = 2c.$$

Note that a, b, c are positive real numbers and they satisfy $ab + bc + ca = abc$. Observe that

$$\begin{aligned}(a-1)(b-1)(c-1) &= abc - ab - bc - ca + a + b + c - 1\\ &= a + b + c - 1,\\ (a+1)(b+1)(c+1) &= abc + ab + bc + ca + a + b + c + 1\\ &= 2abc + a + b + c + 1.\end{aligned}$$

Using the AM-GM-HM inequality, we obtain

$$\frac{a+b+c}{3} \geq (abc)^{\frac{1}{3}} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}},$$

which yields $a + b + c \geq 9$ and $abc \geq 3^3$. This implies that

$$\begin{aligned}&\frac{1}{(a-1)(b-1)(c-1)} + \frac{8}{(a+1)(b+1)(c+1)}\\ &\leq \frac{1}{9-1} + \frac{8}{2 \cdot 27 + 9 + 1}\\ &= \frac{1}{8} + \frac{8}{64}\\ &= \frac{1}{4}.\end{aligned}$$

■

Exercise 47 (ELMO 2013 P2). Let a, b, c be positive reals satisfying $a+b+c = \sqrt[7]{a} + \sqrt[7]{b} + \sqrt[7]{c}$. Prove that $a^a b^b c^c \geq 1$.

Walkthrough —
(a)

Solution 45. Using the weighted AM-GM inequality, we obtain

$$1 = \sum_{\text{cyc}} \frac{a}{a+b+c} \cdot a^{-\frac{6}{7}} \geq (a^a b^b c^c)^{-\frac{6/7}{a+b+c}},$$

which yields $a^a b^b c^c \geq 1$. ■

Exercise 48 (RMO 2014e P5,). Let a, b, c be positive real numbers such that

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \leq 1. \tag{3}$$

Prove that

$$(1+a^2)(1+b^2)(1+c^2) \geq 125.$$

When does the equality hold?

Solution 46. The given inequality implies

$$\frac{a}{1+a} \geq \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{2}{\sqrt{(1+b)(1+c)}},$$

where the second inequality holds if and only if $b = c$. Similar lower bounds for $\frac{b}{1+b}, \frac{c}{1+c}$ can be obtained, and multiplying them yields $abc \geq 8$, where equality holds if and only if $a = b = c = 2$. Note that

$$1 + a^2 = 1 + \frac{a^2}{4} + \frac{a^2}{4} + \frac{a^2}{4} + \frac{a^2}{4} \geq 5 \left(\frac{a^8}{2^8}\right)^{\frac{1}{5}}$$

holds, where equality occurs if and only if $a = 2$. We can find similar lower bounds for $1 + b^2, 1 + c^2$. Multiplying them, we get

$$(1+a^2)(1+b^2)(1+c^2) \geq 5^3 \left(\frac{a^8}{2^8}\frac{b^8}{2^8}\frac{c^8}{2^8}\right)^{\frac{1}{5}} = 5^3 \left(\frac{abc}{8}\right)^{\frac{8}{5}} \geq 125,$$

where equality occurs if and only if $a = b = c = 2$. ■

Exercise 49 (RMO 2016c P2,). Let a, b, c be three distinct positive real numbers such that $abc = 1$. Prove that

$$\frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)} \geq 3.$$

Solution 47. Note that

$$\begin{aligned}
&\frac{a^3}{(a-b)(a-c)}+\frac{b^3}{(b-c)(b-a)}+\frac{c^3}{(c-a)(c-b)}\\
&=-\frac{a^3(b-c)+b^3(c-a)+c^3(a-b)}{(a-b)(b-c)(c-a)}\\
&=-\frac{a^3(b-c)+b^3(c-a)-c^3(b-c)-c^3(c-a)}{(a-b)(b-c)(c-a)}\\
&=-\frac{(b^3-c^3)(c-a)+(a^3-c^3)(b-c)}{(a-b)(b-c)(c-a)}\\
&=-\frac{b^2+bc+c^2-c^2-a^2-ca}{a-b}\\
&=-\frac{b^2+bc-a^2-ca}{a-b}\\
&=a+b+c\\
&\geq 3\sqrt[3]{abc}\\
&=3.
\end{aligned}$$

■

Remark. The above argument leads to following somewhat simpler solution. Observing that

$$\begin{aligned}
\frac{1}{(c-a)(c-b)}&=\frac{(c-a)-(c-b)}{(b-a)(c-a)(c-b)}\\
&=\frac{1}{(b-a)(c-b)}-\frac{1}{(b-a)(c-a)},
\end{aligned}$$

we obtain

$$\begin{aligned}
&\frac{a^3}{(a-b)(a-c)}+\frac{b^3}{(b-c)(b-a)}+\frac{c^3}{(c-a)(c-b)}\\
&=\frac{a^3-c^3}{(a-b)(a-c)}+\frac{b^3-c^3}{(b-c)(b-a)}\\
&=\frac{a^2+ac+c^2}{a-b}-\frac{b^2+bc+c^2}{a-b}\\
&=a+b+c\\
&\geq 3.
\end{aligned}$$

Exercise 50 (RMO 2016e P4,). Let a, b, c be positive real numbers such that $a+b+c=3$. Determine, with certainty, the largest possible value of the expression

$$\frac{a}{a^3+b^2+c}+\frac{b}{b^3+c^2+a}+\frac{c}{c^3+a^2+b}.$$

Solution 48. Note that

$$\begin{aligned}
&\frac{a}{a^3+b^2+c}+\frac{b}{b^3+c^2+a}+\frac{c}{c^3+a^2+b}\\
&=\frac{1}{a^2+\frac{b^2}{a}+\frac{c}{a}}+\frac{1}{b^2+\frac{c^2}{b}+\frac{a}{b}}+\frac{1}{c^2+\frac{a^2}{c}+\frac{b}{c}}\\
&\le\frac{1+a+ca}{(a+b+c)^2}+\frac{1+b+ab}{(a+b+c)^2}+\frac{1+c+bc}{(a+b+c)^2}\\
&\qquad\text{(by the Cauchy–Schwarz inequality)}\\
&=\frac{3+a+b+c+ab+bc+ca}{(a+b+c)^2}\\
&\le\frac{6+\frac{(a+b+c)^2}{3}}{9}\\
&=1.
\end{aligned}$$

Also note that if $a=b=c=1$, then $a+b+c=3$ and

$$\frac{a}{a^3+b^2+c}+\frac{b}{b^3+c^2+a}+\frac{c}{c^3+a^2+b}=1.$$

This shows that the largest possible value of the given expression is equal to 1. ■

Exercise 51 (RMO 2016f P5,). Let x,y,z be non-negative real numbers such that $xyz=1$. Prove that

$$(x^3+2y)(y^3+2z)(z^3+2x)\ge 27.$$

Solution 49. Applying the AM-GM inequality, we obtain

$$\begin{aligned}
&(x^3+2y)(y^3+2z)(z^3+2x)\\
&\ge(3\sqrt[3]{x^3y^2})(3\sqrt[3]{y^3z^2})(3\sqrt[3]{z^3x^2})\\
&=27.
\end{aligned}$$

■

Exercise 52 (RMO 2019a P3,). Let a,b,c be positive real numbers such that $a+b+c=1$. Prove that

$$\frac{a}{a^2+b^3+c^3}+\frac{b}{b^2+a^3+c^3}+\frac{c}{c^2+a^3+b^3}\le\frac{1}{5abc}.$$

Walkthrough — **Homogenize** the denominator and apply the AM-GM inequality.

Solution 50. Note that

$$\begin{aligned}\frac{a}{a^2+b^3+c^3} &= \frac{a}{a^2(a+b+c)+b^3+c^3}\\ &\le \frac{a}{5\sqrt[5]{a^6\cdot abc\cdot b^3\cdot c^3}}\\ &= \frac{1}{5\sqrt[5]{a^2b^4c^4}}\\ &= \frac{1}{5abc}\left(a^3bc\right)^{1/5}\\ &\le \frac{3a+b+c}{25abc}\end{aligned}$$

holds. Similarly, it follows that

$$\begin{aligned}\frac{b}{b^2+a^3+c^3} &\le \frac{3b+a+c}{25abc},\\ \frac{c}{c^2+a^3+b^3} &\le \frac{3c+a+b}{25abc}.\end{aligned}$$

Adding the above inequalities, we obtain

$$\begin{aligned}&\frac{a}{a^2+b^3+c^3}+\frac{b}{b^2+a^3+c^3}+\frac{c}{c^2+a^3+b^3}\\ &\le \frac{3a+b+c}{25abc}+\frac{3b+a+c}{25abc}+\frac{3c+a+b}{25abc}\\ &= \frac{a+b+c}{5abc}\\ &= \frac{1}{5abc}.\end{aligned}$$

■

Exercise 53 (Canadian Mathematical Olympiad 2021 P2,). Let $n \ge 2$ be some fixed positive integer and suppose that $a_1, a_2, \ldots, a_n$ are positive real numbers satisfying

$$a_1 + a_2 + \cdots + a_n = 2^n - 1.$$

Find the minimum possible value of

$$\frac{a_1}{1} + \frac{a_2}{1+a_1} + \frac{a_3}{1+a_1+a_2} + \cdots + \frac{a_n}{1+a_1+a_2+\cdots+a_{n-1}}.$$

Walkthrough —
(a)

Solution 51. Note that

$$\begin{aligned}
&\frac{a_1}{1}+\frac{a_2}{1+a_1}+\frac{a_3}{1+a_1+a_2}+\cdots+\frac{a_n}{1+a_1+a_2+\cdots+a_{n-1}}\\
&=\frac{1+a_1}{1}+\frac{1+a_1+a_2}{1+a_1}+\frac{1+a_1+a_2+a_3}{1+a_1+a_2}\\
&\quad+\cdots+\frac{1+a_1+a_2+\cdots+a_n}{1+a_1+a_2+\cdots+a_{n-1}}-n\\
&\geq n\sqrt[n]{1+a_1+a_2+\cdots+a_n}-n\\
&=n\sqrt[n]{2^n}-n\\
&=n.
\end{aligned}$$

This shows that the minimum possible value of the given expression is at least n.

Also note that if $a_i=2^{i-1}$ for all $1\leq i\leq n$, then $a_1+a_2+\cdots+a_n=2^n-1$ holds, and

$$\begin{aligned}
&\frac{a_1}{1}+\frac{a_2}{1+a_1}+\frac{a_3}{1+a_1+a_2}+\cdots+\frac{a_n}{1+a_1+a_2+\cdots+a_{n-1}}\\
&=\frac{1+a_1}{1}+\frac{1+a_1+a_2}{1+a_1}+\frac{1+a_1+a_2+a_3}{1+a_1+a_2}\\
&\quad+\cdots+\frac{1+a_1+a_2+\cdots+a_n}{1+a_1+a_2+\cdots+a_{n-1}}-n\\
&=2n-n\\
&=n.
\end{aligned}$$

This proves that the minimum possible value of the given expression is equal to n. ■

§1.7 Bunching terms

Example 54. Show that

$$\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{2^m}\geq\frac{m}{2}\quad\text{for } m\geq 1.$$

Use it to show that $\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{n}$ becomes arbitrarily large as n increases indefinitely.

The content posted here and at this blog by Evan Chen are quite useful.

Walkthrough — Observe that

$$\begin{aligned}&\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{2^m}\\&=\frac{1}{2}\\&\quad+\left(\frac{1}{3}+\frac{1}{4}\right)\\&\quad+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right)\\&\quad+\dots\\&\quad+\left(\frac{1}{2^{m-1}}+\frac{1}{2^{m-1}+1}+\cdots+\frac{1}{2^m}\right).\end{aligned}$$

Show that the sum in each parenthesis is at most $\frac{1}{2}$.

Exercise 55 (RMO 1992 P7,). Show that

$$1<\frac{1}{1001}+\frac{1}{1002}+\cdots+\frac{1}{3001}<\frac{4}{3}$$

Solution 52. To obtain the lower bound, note that

$$\begin{aligned}&\frac{1}{1001}+\frac{1}{1002}+\cdots+\frac{1}{3001}\\&>\frac{1}{1001}+\frac{1}{1002}+\cdots+\frac{1}{3000}\\&=\left(\frac{1}{1001}+\cdots+\frac{1}{1250}\right)+\left(\frac{1}{1251}+\cdots+\frac{1}{1500}\right)+\left(\frac{1}{1501}+\cdots+\frac{1}{1750}\right)\\&+\left(\frac{1}{1751}+\cdots+\frac{1}{2000}\right)+\left(\frac{1}{2001}+\cdots+\frac{1}{2250}\right)+\left(\frac{1}{2251}+\cdots+\frac{1}{2500}\right)\\&+\left(\frac{1}{2501}+\cdots+\frac{1}{2750}\right)+\left(\frac{1}{2751}+\cdots+\frac{1}{3000}\right)\\&>\frac{250}{1250}+\frac{250}{1500}+\frac{250}{1750}+\frac{250}{2000}+\frac{250}{2250}+\frac{250}{2500}+\frac{250}{2750}+\frac{250}{3000}\\&>\frac{1}{5}+\frac{1}{6}+\cdots+\frac{1}{12}=17\left(\frac{1}{60}+\frac{1}{66}+\frac{1}{70}+\frac{1}{72}\right)>1.\end{aligned}$$

To get the upper bound, note that

$$\begin{aligned}&\frac{1}{1001}+\frac{1}{1002}+\cdots+\frac{1}{3001}\\&<\frac{1}{1000}+\frac{1}{1001}+\cdots+\frac{1}{3000}\\&<\left(\frac{1}{1000}+\frac{1}{1001}+\cdots+\frac{1}{1499}\right)+\left(\frac{1}{1500}+\frac{1}{1501}+\cdots+\frac{1}{1999}\right)\end{aligned}$$

$$
\begin{aligned}
&+\left(\frac{1}{2000}+\frac{1}{2001}+\cdots+\frac{1}{2499}\right)+\left(\frac{1}{2500}+\frac{1}{2501}+\cdots+\frac{1}{2999}\right)+\frac{1}{3000}\\
&<\frac{500}{1500}+\frac{500}{2000}+\frac{500}{2500}+\frac{500}{3000}+\frac{1}{3000}\\
&=\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{3000}\\
&=\frac{57}{60}+\frac{1}{3000}\\
&<1\\
&<\frac{4}{3}.
\end{aligned}
$$

■

Exercise 56 (RMO 1998 P3, , Canadian Mathematical Olympiad 1998 P3)**.** Prove the following inequality for every natural number $n \geq 2$:

$$\frac{1}{n+1}\left(1+\frac{1}{3}+\frac{1}{5}+\cdots+\frac{1}{2n-1}\right)>\frac{1}{n}\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{6}+\cdots+\frac{1}{2n}\right).$$

Solution 53. Let us establish the following Claim.

> **Claim —** For any $1 \leq k \leq n$, the inequality
>
> $$\frac{1}{n+1}\left(\frac{1}{2k-1}+\frac{1}{2n-2k+1}\right)\geq\frac{1}{n}\left(\frac{1}{2k}+\frac{1}{2n-2k+2}\right)$$
>
> holds. Moreover, this inequality is strict if $k=1$ and $n \geq 2$.

Proof of the Claim. Clearing the denominators, it follows that the above inequality is equivalent to

$$4n^2k(n-k+1)\geq(n+1)^2(2k-1)(2n-2k+1).$$

Note that

$$
\begin{aligned}
&4n^2k(n-k+1)-(n+1)^2(2k-1)(2n-2k+1)\\
&=n^2(4nk-4k^2+4k-(4nk-4k^2+4k-2n-1))\\
&\quad-(2n+1)(2k-1)(2n-2k+1)\\
&=n^2(2n+1)-(2n+1)(2k-1)(2n-2k+1)\\
&=(2n+1)(n^2-(4nk-4k^2+4k-2n-1))\\
&=(2n+1)((n-2k)^2+2(n-2k)+1)\\
&=(2n+1)(n-2k+1)^2,
\end{aligned}
$$

which is nonnegative for any $1 \leq k \leq n$. This proves the inequality. Moreover, if $k = 1$ and $n \geq 2$, then

$$4n^2k(n-k+1) - (n+1)^2(2k-1)(2n-2k+1) > 0,$$

which proves the second statement. □

The Claim implies the given inequality. ■

§1.8 Look at the difference

Example 57. Show that if $a > b > 0$, then

$$a + \frac{1}{(a-b)b} \geq 3.$$

Solution 54. Applying the AM-GM inequality, we obtain

$$a + \frac{1}{(a-b)b} = (a-b) + b + \frac{1}{(a-b)b} \geq 3.$$

■

Example 58. Prove that if $a < b < c < d$, then $(a+b+c+d)^2 > 8(ac+bd)$.

Solution 55. Let x, y, z denote the real numbers satisfying

$$\begin{aligned} b &= a + x, \\ c &= b + y, \\ d &= c + z. \end{aligned}$$

Note that

$$\begin{aligned} a+b+c+d &= a+b+2c+(d-c) \\ &= a+3b+2(c-b)+(d-c) \\ &= 4a+3(b-a)+2(c-b)+(d-c) \\ &= 4a+3x+2y+z, \\ ac+bd &= ac+(a+x)(c+z) \\ &= 2ac+cx+az+zx \\ &= 2a(b+y)+x(b+y)+az+zx \\ &= 2a(a+x+y)+x(a+x+y)+az+zx \\ &= 2a^2+2ax+2ay+ax+x^2+xy+az+zx \\ &= 2a^2+3ax+2ay+zx+x^2+xy+az \end{aligned}$$

hold. This shows that

$$\begin{aligned}
&(a+b+c+d)^2 - 8(ac+bd)\\
&= (4a+3x+2y+z)^2 - 8(2a^2+3ax+2ay+zx+x^2+xy+az)\\
&= 9x^2+4y^2+z^2+24ax+16ay+8az+12xy+4yz+6zx\\
&\quad -24ax-16ay-8zx-8x^2-8xy-8az\\
&= x^2+4y^2+z^2+4xy+4yz-2zx\\
&= (x-z)^2+4y(x+y+z)\\
&> 0.
\end{aligned}$$

■

Exercise 59 (RMO 2004 P7,). Let x and y be positive real numbers such that $y^3+y \le x-x^3$. Prove that

1. $y < x < 1$ and
2. $x^2+y^2 < 1$.

Solution 56. Note that $x-y \ge x^3+y^3 > 0$ holds, which shows that $x > y$. Also note that $x-x^3 \ge y^3+y > 0$ holds, which implies that $x(1-x^2) > 0$. Since x is positive, we obtain $x < 1$. This gives $0 < y < x < 1$.

Write $x = y+t$ with $t > 0$. The inequality $y^3+y \le x-x^3$ yields

$$2y^3+3y^2t+3yt+t^3 \le t.$$

Note that the inequality $x^2+y^2 < 1$ is equivalent to $2y^2+2yt+t^2 < 1$. Since $y+t$ is positive, it is equivalent to $(y+t)(2y^2+2yt+t^2) < y+t$. Observe that

$$\begin{aligned}
(y+t)(2y^2+2yt+t^2) &= 2y^3+4y^2t+3yt^2+t^3\\
&\le y^2t+3yt^2+t-3yt\\
&= y^2t-3yt(1-t)+t\\
&< y+t \quad (\text{using } 0<y,t<1).
\end{aligned}$$

This completes the proof. ■

Exercise 60 (RMO 2017b P5). If $a,b,c,d \in \mathbb{R}$ such that $a > b > c > d > 0$ and $a+d = b+c$; then prove that

$$\frac{(a+b)-(c+d)}{\sqrt{2}} > \sqrt{a^2+b^2}-\sqrt{c^2+d^2}.$$

The content posted here and at this blog by Evan Chen are quite useful.

Solution 57. Put $m = a + d$. Using $a + d = b + c$ and $a > b > c > d > 0$, it follows that for some real numbers $0 < y < x < m$,

$$a = m + x, b = m + y, c = m - y, d = m - x$$

hold. Note that the given inequality reduces to

$$\sqrt{2}(x + y) > \sqrt{(m + x)^2 + (m + y)^2} - \sqrt{(m - x)^2 + (m - y)^2}.$$

Observe that

$$\begin{aligned}
&\sqrt{(m + x)^2 + (m + y)^2} - \sqrt{(m - x)^2 + (m - y)^2} \\
&= \frac{(m + x)^2 + (m + y)^2 - (m - x)^2 - (m - y)^2}{\sqrt{(m + x)^2 + (m + y)^2} + \sqrt{(m - x)^2 + (m - y)^2}} \\
&= \frac{4m(x + y)}{\sqrt{(m + x)^2 + (m + y)^2} + \sqrt{(m - x)^2 + (m - y)^2}}.
\end{aligned}$$

Hence, it suffices to prove that

$$\sqrt{(m + x)^2 + (m + y)^2} + \sqrt{(m - x)^2 + (m - y)^2} > 2\sqrt{2}m.$$

Note that

$$\begin{aligned}
&\sqrt{(m + x)^2 + (m + y)^2} + \sqrt{(m - x)^2 + (m - y)^2} \\
&> \frac{m + x + m + y}{\sqrt{2}} + \frac{m - x + m - y}{\sqrt{2}} \\
&= 2\sqrt{2}m
\end{aligned}$$

hold. This completes the proof. ■

Exercise 61 (RMO 2024b P4,). Let a_1, a_2, a_3, a_4 be real numbers such that $a_1^2 + a_2^2 + a_3^2 + a_4^2 = 1$. Show that there exist i, j with $1 \leq i < j \leq 4$, such that $(a_i - a_j)^2 \leq \frac{1}{5}$.

Solution 58. Reordering a_1, a_2, a_3, a_4 if necessary, we assume that $a_1 \leq a_2 \leq a_3 \leq a_4$ holds. Put

$$\begin{aligned}
a_2 &= a_1 + x, \\
a_3 &= a_2 + y, \\
a_4 &= a_3 + z.
\end{aligned}$$

Note that

$$\begin{aligned}
a_2^2 &= a_1^2 + x^2 + 2a_1x, \\
a_3^2 &= (a_1 + x + y)^2
\end{aligned}$$

$$
\begin{aligned}
&= a_1^2 + x^2 + y^2 + 2a_1x + 2a_1y + 2xy, \\
a_4^2 &= (a_1 + x + y + z)^2 \\
&= a_1^2 + x^2 + y^2 + z^2 + 2a_1x + 2a_1y + 2a_1z + 2xy + 2yz + 2zx
\end{aligned}
$$

hold. This gives

$$
\begin{aligned}
&a_1^2 + a_2^2 + a_3^2 + a_4^2 \\
&= a_1^2 \\
&\quad + a_1^2 + x^2 + 2a_1x \\
&\quad + a_1^2 + x^2 + y^2 + 2a_1x + 2a_1y + 2xy \\
&\quad + a_1^2 + x^2 + y^2 + z^2 + 2a_1x + 2a_1y + 2a_1z + 2xy + 2yz + 2zx \\
&= 4a_1^2 + 3x^2 + 2y^2 + z^2 + 2a_1(3x + 2y + z) + 4xy + 2yz + 2zx \\
&= \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 - \frac{(3x + 2y + z)^2}{4} \\
&\quad + 3x^2 + 2y^2 + z^2 + 4xy + 2yz + 2zx \\
&= \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 - \left(\frac{9}{4}x^2 + y^2 + \frac{1}{4}z^2 + 3xy + yz + \frac{3}{2}zx\right) \\
&\quad + 3x^2 + 2y^2 + z^2 + 4xy + 2yz + 2zx \\
&= \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 + \frac{3}{4}x^2 + y^2 + \frac{3}{4}z^2 + xy + yz + \frac{1}{2}zx.
\end{aligned}
$$

If $x > \frac{1}{\sqrt{5}}, y > \frac{1}{\sqrt{5}}, z > \frac{1}{\sqrt{5}}$ holds, then we would obtain

$$
\begin{aligned}
1 &> \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 + \left(\frac{3}{4} + 1 + \frac{3}{4} + 1 + 1 + \frac{1}{2}\right)\frac{1}{5} \\
&= \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 + 1 \\
&\geq 1,
\end{aligned}
$$

which yields $1 > 1$, which is impossible. This shows that at least one of x, y, z is less than or equal to $\frac{1}{\sqrt{5}}$. ■

§1.9 Algebraic substitutions

Exercise 62 (RMO 2016a P2, , RMO 2016b P2)**.** Let a, b, c be positive real numbers such that

$$\frac{a}{1+a} + \frac{b}{1+b} + \frac{c}{1+c} = 1.$$

Prove that $abc \leq \frac{1}{8}$.

Solution 59. Put

$$x = \frac{a}{1+a}, y = \frac{b}{1+b}, z = \frac{c}{1+c}.$$

It follows that x, y, z are positive real numbers satisfying $x + y + z = 1$. Note that

$$a = \frac{x}{1-x}, b = \frac{y}{1-y}, c = \frac{z}{1-z}.$$

We need to shows that $8abc \leq 1$, which is equivalent to $8xyz \leq (1-x)(1-y)(1-z)$. Using $x+y+z = 1$, it follows that the above inequality is equivalent to

$$8xyz \leq (x+y)(y+z)(z+x),$$

which follows from the AM-GM inequality. ■

> **Remark.** A careful reading of the above argument shows that the substitution
>
> $$\frac{x}{x+y+z} = \frac{a}{1+a}, \frac{y}{x+y+z} = \frac{b}{1+b}, \frac{z}{x+y+z} = \frac{c}{1+c},$$
>
> or equivalently, the substitution
>
> $$a = \frac{x}{y+z}, b = \frac{y}{z+x}, c = \frac{z}{x+y}$$
>
> reduces $8abc \leq 1$ to the inequality $8xyz \leq (x+y)(y+z)(z+x)$.

We refer to Exercise 29.

Exercise 63 (RMO 2016b P2, , RMO 2016a P2)**.** Let a, b, c be positive real numbers such that

$$\frac{a}{1+a} + \frac{b}{1+b} + \frac{c}{1+c} = 1.$$

Prove that $abc \leq \frac{1}{8}$.

Solution 60. Put

$$x = \frac{a}{1+a}, y = \frac{b}{1+b}, z = \frac{c}{1+c}.$$

It follows that x, y, z are positive real numbers satisfying $x + y + z = 1$. Note that

$$a = \frac{x}{1-x}, b = \frac{y}{1-y}, c = \frac{z}{1-z}.$$

We need to shows that $8abc \leq 1$, which is equivalent to $8xyz \leq (1-x)(1-y)(1-z)$. Using $x+y+z = 1$, it follows that the above inequality is equivalent to

$$8xyz \leq (x+y)(y+z)(z+x),$$

which follows from the AM-GM inequality. ■

Remark. A careful reading of the above argument shows that the substitution

$$\frac{x}{x+y+z}=\frac{a}{1+a}, \frac{y}{x+y+z}=\frac{b}{1+b}, \frac{z}{x+y+z}=\frac{c}{1+c},$$

or equivalently, the substitution

$$a=\frac{x}{y+z}, b=\frac{y}{z+x}, c=\frac{z}{x+y}$$

reduces $8abc \leq 1$ to the inequality $8xyz \leq (x+y)(y+z)(z+x)$.

We refer to Exercise 29.

Exercise 64 (RMO 2025a P2,). Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that

$$\frac{2a^2}{a^3+1}+\frac{2b^2}{b^3+1}+\frac{2c^2}{c^3+1} \leq a^2+b^2+c^2.$$

Walkthrough —

(a)

Solution 61.

■

§1.10 Ravi substitution

In Fig. 1, the incircle touches the sides of the triangle ABC at the points D, E, F. Note that $AF = AE, BF = BD, CD = CE$ holds, and let us denote them by x, y, z respectively. This gives

$$a = BC = y + z, b = CA = z + x, c = AB = x + y$$

with $x, y, z > 0$.

Exercise 65 (RMO 1996 P5,). Let ABC be a triangle and h_a the altitude through A. Prove that

$$(b+c)^2 \geq a^2 + 4h_a^2.$$

Solution 62. Since $\Delta = \frac{1}{2}ah_a$, the inequality reduces to

$$(b+c)^2 - a^2 \geq \frac{1}{a^2}16\Delta^2,$$

which is equivalent to

$$a^2 \geq (c+a-b)(a+b-c).$$

This follows from the AM-GM inequality. ■

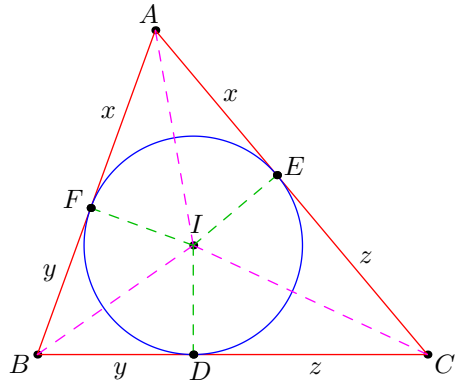


Figure 1: Ravi substitution

Remark. To prove the inequality $a^2 \geq (c+a-b)(a+b-c)$, one could substitute

$$a = y + z, b = z + x, c = x + y,$$

and note that the above inequality is equivalent to

$$(y + z)^2 \geq 4yz,$$

which holds since $(y - z)^2 \geq 0$.

Exercise 66 (RMO 1999 P5,). If a, b, c are the sides of a triangle, prove the following inequality:

$$\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \geq 3.$$

Solution 63. Let us use Ravi substitution, that is, let x, y, z be real numbers satisfying

$$a = y + z, b = z + x, c = x + y.$$

Note that x, y, z are positive by the triangle inequality. Observe that

$$\begin{aligned}
&\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \\
&= \frac{y+z}{2y} + \frac{z+x}{2z} + \frac{x+y}{2x} \\
&= \frac{3}{2} + \frac{1}{2}\left(\frac{y}{x} + \frac{z}{y} + \frac{x}{z}\right) \\
&\geq \frac{3}{2} + \frac{3}{2} \quad \text{(by the AM-GM inequality)} \\
&= 3.
\end{aligned}$$

■

Exercise 67 (RMO 2001 P6,). If x, y, z are the sides of a triangle, then prove that

$$|x^2(y-z)+y^2(z-x)+z^2(x-y)|<xyz.$$

Solution 64. Note that

$$\begin{aligned}
&x^2(y-z)+y^2(z-x)+z^2(x-y)\\
&=x^2y-zx^2+y^2z-xy^2+z^2x-yz^2\\
&=(x^2-z^2)y+y^2(z-x)+zx(z-x)\\
&=(z-x)(y^2+zx-y(z+x))\\
&=(z-x)(y-z)(y-x).
\end{aligned}$$

Since

$$|x^2(y-z)+y^2(z-x)+z^2(x-y)|, xyz$$

are symmetric in x, y, z, without loss of generality, we may (and do) assume that $x \le y \le z$. Note that $|(z-x)(y-z)(y-x)|<xyz$ is immediate if any two of x, y, z are equal. It remains to consider the case that x, y, z are distinct, which we assume from now on. Let us use Ravi substitution, that is, let a, b, c be real numbers satisfying

$$x=b+c, y=c+a, z=a+b,$$

where a, b, c are positive by the triangle inequality. Using $x<y<z$, we obtain $a>b>c$. Note that

$$\begin{aligned}
|(z-x)(y-z)(y-x)| &= (a-b)(b-c)(a-c)\\
&< aba\\
&< (a+c)(b+c)(a+b)\\
&= xyz,
\end{aligned}$$

where the first inequality follows since $0<a-b<a, 0<b-c<b, 0<a-c<c$ holds. ■

§1.11 Triangle inequality

Example 68. For real numbers x, y, z, show that

$$|x|+|y|+|z| \le |x+y-z|+|y+z-x|+|z+x-y|.$$

Solution 65. Put

$$x=\frac{b+c}{2}, y=\frac{c+a}{2}, z=\frac{a+b}{2}.$$

The content posted here and at this blog by Evan Chen are quite useful.

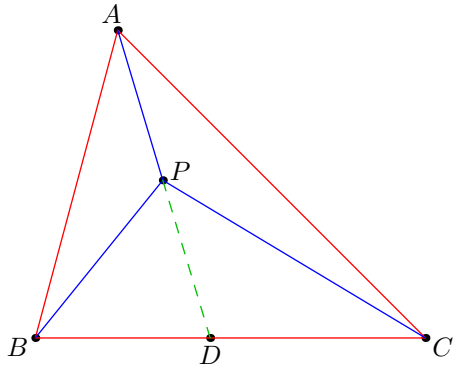


Figure 2: India RMO 1990 P5, Exercise 69

In other words, let a, b, c be real numbers defined by

$$a = y + z - x, b = z + x - y, c = x + y - z.$$

Applying the triangle inequality, we obtain

$$|a| + |b| \geq 2|z|, |b| + |c| \geq 2|x|, |c| + |a| \geq 2|y|.$$

The result follows by adding them. ■

Exercise 69 (RMO 1990 P5,). Let P be a point inside a triangle ABC. Let s denote the semiperimeter $\frac{1}{2}(AB + BC + CA)$. Prove that

$$s < AP + BP + CP < 2s.$$

Solution 66. Applying the triangle inequality to PBC, we get $BC < BP+CP$. Similarly, the inequalities $CA < CP + AP, AB < AP + BP$ follow. Adding them yields

$$AB + BC + CA < 2(AP + BP + CP),$$

or equivalently, $s < AP + BP + CP$ holds. Let D denote the point on BC such that A, P, D are collinear. Note that

$$\begin{aligned} AP + BP &< AP + BD + DP && \text{(applying the triangle inequality to } BDP) \\ &= AD + BD \\ &< AC + DC + BD && \text{(applying the triangle inequality to } ADC) \\ &= AC + BC \end{aligned}$$

hold. Similarly, it follows that

$$BP + CP < AB + CA, CP + AP < AB + BC.$$

Adding these inequalities, we obtain

$$AP + BP + CP < AB + BC + CA = 2s.$$

■

Exercise 70 (RMO 1995 P5,). Show that for any triangle ABC, the following inequality is true:

$$a^2 + b^2 + c^2 > \sqrt{3}\max\{|a^2 - b^2|, |b^2 - c^2|, |c^2 - a^2|\}.$$

Solution 67. Since the above inequality is symmetric with respect to a, b, c, without loss of generality, we assume that $a > b > c$. Thus we need to show that

$$a^2 + b^2 + c^2 > \sqrt{3}(a^2 - c^2)$$

holds. Using the triangle inequality, we obtain $b > a - c > 0$. This yields

$$\begin{aligned} a^2 + b^2 + c^2 - \sqrt{3}(a^2 - c^2) &> a^2 + (a-c)^2 + c^2 - \sqrt{3}(a^2 - c^2) \\ &= 2(a^2 + c^2 - ca) - \sqrt{3}(a^2 - c^2) \\ &= (2 - \sqrt{3})a^2 + (2 + \sqrt{3})c^2 - 2ca \\ &= (2 - \sqrt{3})(a^2 - 2ac(2 + \sqrt{3}) + (2 + \sqrt{3})^2) \\ &= (2 - \sqrt{3})(a - (2 + \sqrt{3})c)^2 \\ &\geq 0. \end{aligned}$$

This completes the proof. ■

Exercise 71 (RMO 1997 P5,). Let x, y and z be three distinct real positive numbers. Determine with proof whether or not the three real numbers

$$\left|\frac{x}{y} - \frac{y}{x}\right|, \left|\frac{y}{z} - \frac{z}{y}\right|, \left|\frac{z}{x} - \frac{x}{z}\right|$$

can be the lengths of the sides of a triangle.

Solution 68. Without loss of generality, let us assume that $x > y > z$. Note that

$$\begin{aligned} &z(x^2 - y^2) + x(y^2 - z^2) - y(x^2 - z^2) \\ &= zx(x - z) + (x - z)y^2 - y(x^2 - z^2) \\ &= (x - z)(zx + y^2 - y(x + z)) \\ &= (x - z)(y - x)(y - z) \\ &< 0. \end{aligned}$$

By the triangle inequality, it follows that the given three numbers cannot be the lengths of the sides of a triangle. ■

Exercise 72 (RMO 2009 P5,). A convex polygon is such that the distance between any two vertices does not exceed 1.

(i) Prove that the distance between any two points on the boundary of the polygon does not exceed 1.

(ii) If X and Y are two distinct points inside the polygon, prove that there exists a point Z on the boundary of the polygon such that $XZ+YZ \leq 1$.

Solution 69. Note that for any four complex numbers z_1, z_2, z_3, z_4 and any $0 \leq t, s \leq 1$, we have

$$\begin{aligned}&(tz_1+(1-t)z_2)-(sz_3+(1-s)z_4)\\&=t(z_1-z_3)+(t-s)z_3+(1-s)(z_2-z_4)+(s-t)z_2\\&=t(z_1-z_3)+(s-t)(z_3-z_2)+(1-s)(z_2-z_4)\end{aligned}$$

and

$$\begin{aligned}&(tz_1+(1-t)z_2)-(sz_3+(1-s)z_4)\\&=s(z_1-z_3)+(t-s)z_1+(1-t)(z_2-z_4)+(s-t)z_4\\&=s(z_1-z_3)+(t-s)(z_1-z_4)+(1-t)(z_2-z_4).\end{aligned}$$

If the distance between no two of z_1, z_2, z_3, z_4 exceeds 1, then

$$|(tz_1+(1-t)z_2)-(sz_3+(1-s)z_4)| \leq \begin{cases} t+(s-t)+(1-s)=1 & \text{if } s \geq t,\\ s+(t-s)+(1-t)=1 & \text{if } t \geq s.\end{cases}$$

This proves part (i).

Extending XY to the boundary of the polygon, we find two points Z_1, Z_2 on the boundary such that Z_1, X, Y, Z_2 are collinear. Note that

$$\begin{aligned}(XZ_1+YZ_1)+(XZ_2+YZ_2)&=(XZ_1+XZ_2)+(YZ_1+YZ_2)\\&=2Z_1Z_2\\&\leq 2,\end{aligned}$$

which shows that $XZ_1+YZ_1 \leq 1$ or $XZ_2+YZ_2 \leq 1$ holds. This proves part (ii). ■

Exercise 73 (RMO 2014e P1,). Three positive real numbers a, b, c are such that $a^2+5b^2+4c^2-4ab-4bc=0$. Can a, b, c be the lengths of the sides of a triangle? Justify your answer.

Solution 70. The given condition is equivalent to

$$(a-2b)^2+(b-2c)^2=0.$$

Since a, b, c are real numbers, this gives $b=2c, a=2b=4c$. Note that $b+c=3c<4c=a$ holds. So by the triangle inequality, it follows that a, b, c cannot be the lengths of the sides of a triangle. ■

Exercise 74 (IOQM 2024 P10,). Determine the number of positive integral values of p for which there exists a triangle with sides a, b and c which satisfy

$$a^2 + (p^2+9)b^2 + 9c^2 - 6ab - 6pbc = 0$$

Summary — Complete squares and use the triangle inequality.

Walkthrough —

(a) Note that

$$a^2 + (p^2+9)b^2 + 9c^2 - 6ab - 6pbc = (a-3b)^2 + (pb-3c)^2.$$

(b) This gives $a = 3b$ and

$$c = \frac{p}{3}b.$$

(c) Use the inequality[a] $a + b > c$, to obtain

$$3b + b > \frac{p}{3}b,$$

which yields $p \leq 11$.

(d) Use the inequality[b] $b + c > a$, to get

$$b + \frac{p}{3}b > 3b,$$

implying $p \geq 7$.

(e) It reduces to finding the number of positive integers p such that $7 \leq p \leq 11$.

What about the condition $c + a > b$? Anything else to be taken care of?

[a] How does it follow?
[b] How does it follow?

Solution 71. ■

§1.12 Geometric inequalities

Exercise 75 (RMO 1993 P4,). Let $ABCD$ be a rectangle with $AB = a$ and $BC = b$. Suppose r_1 is the radius of the circle passing through A and B and touching CD; and similarly r_2 is the radius of the circle passing through B and C and touching AD. Show that

$$r_1 + r_2 \geq \frac{5}{8}(a+b).$$

The content posted here and at this blog by Evan Chen are quite useful.

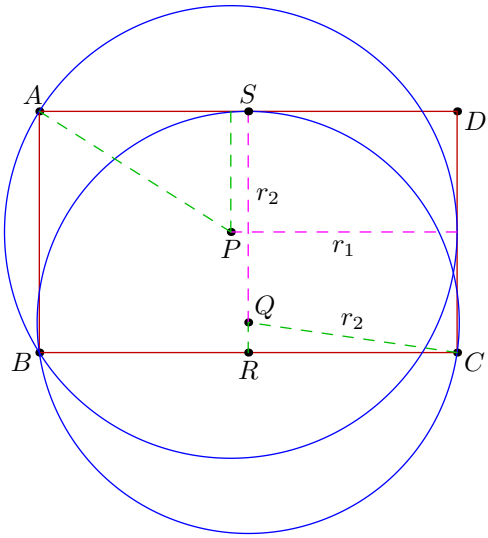


Figure 3: India RMO 1993 P4, Exercise 75

Solution 72. Since QS is perpendicular to AD, it is parallel to CD. Let SQ produced meet BC at R. Since RS is parallel to CD and $\angle RCD$ is a right-angle, it follows that QR is perpendicular to BC. This gives $QR^2 + RC^2 = QC^2$, which yields

$$(a - r_2)^2 + \left(\frac{b}{2}\right)^2 = r_2^2.$$

This implies that

$$r_2 = \frac{a}{2} + \frac{b^2}{8a}.$$

A similar argument shows that

$$r_1 = \frac{b}{2} + \frac{a^2}{8b}.$$

It follows that

$$\begin{aligned} r_1 + r_2 &= \frac{1}{2}(a+b) + \frac{1}{8}\left(\frac{a^2}{b} + \frac{b^2}{a}\right) \\ &\geq \frac{1}{2}(a+b) + \frac{1}{8}\frac{(a+b)^2}{a+b} \quad \text{(by the Cauchy–Schwarz inequality)} \\ &= \frac{5}{8}(a+b). \end{aligned}$$

This completed the proof. ■

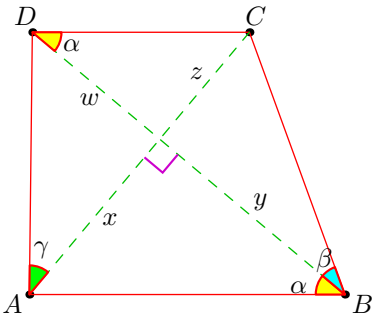


Figure 4: India RMO 1997 P4, Exercise 76

Exercise 76 (RMO 1997 P4,). In a quadrilateral $ABCD$, it is given that AB is parallel to CD and the diagonals AC and BD are perpendicular to each other. Show that

(a) $AD \cdot BC \geq AB \cdot CD$,

(b) $AD + BC \geq AB + CD$.

Solution 73. Note that the inequality $AD \cdot BC \geq AB \cdot CD$ is equivalent to

$$(x^2 + w^2)(y^2 + z^2) \geq (x^2 + y^2)(z^2 + w^2),$$

which reduces to $(x^2-z^2)(y^2-w^2) \geq 0$. It suffices to show that $(x-z)(y-w) \geq 0$ holds. Since AB is parallel to CD, it follows that the triangles ABP and CDP are similar, and the lengths of their sides satisfy

$$\frac{x}{z} = \frac{y}{w}.$$

Consequently, the inequality $(x - z)(y - w) \geq 0$ holds.

To prove the second inequality, note that

$$\begin{aligned}(AD + BC)^2 &= AD^2 + BC^2 + 2AD \cdot BC \\ &= x^2 + w^2 + y^2 + z^2 + 2AD \cdot BC \\ &\geq x^2 + y^2 + z^2 + w^2 + 2AB \cdot CD \\ &= AB^2 + CD^2 + 2AB \cdot CD \\ &= (AB + CD)^2,\end{aligned}$$

which implies $AD + BC \geq AB + CD$. ■

Exercise 77 (RMO 2013b P5,). Let $n \geq 3$ be a natural number and let P be a polygon with n sides. Let $a_1, a_2, \ldots, a_n$ be the lengths of sides of P and let p be its perimeter. Prove that

$$\frac{a_1}{p - a_1} + \frac{a_2}{p - a_2} + \cdots + \frac{a_n}{p - a_n} < 2.$$

Solution 74. By a repeated application of the triangle inequality, it follows that $a_i < p - a_i$ for all $1 \leq i \leq n$. Note that for any positive real numbers a, b, c, the inequality

$$\frac{a}{b} < \frac{a+c}{b+c}$$

holds. It follows that

$$\frac{a_1}{p-a_1} + \frac{a_2}{p-a_2} + \cdots + \frac{a_n}{p-a_n} < \frac{2a_1}{p} + \frac{2a_2}{p} + \cdots + \frac{2a_n}{p} = 2$$

holds. ■

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